

Sean Lange

Full-Stack Software Engineer

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CERTIFICATIONS & EDUCATION

University of South Florida

Bachelor of Science, Computer Science

Tampa, FL

Expected May 2028

IBM Certified Professional Developer – Cloud v6

Cloud Infrastructure, Deployment Pipelines, Distributed Systems

Orlando, FL

October 2025

PROJECTS

Multyx.io | Typescript, Next.js, Tailwind, PostgreSQL, AWS (15 services), Redis, Docker, WebSocket, tRPC

February 2025 - Present

Founded a browser-based multiplayer game development platform enabling beginner programmers to build, test, and deploy real-time games without networking knowledge, currently running closed-beta with 8 testers.

- Designed a **typed AST model** that compiles into **Typescript**, preventing invalid game logic before testing or deployment.
- Configured **AWS ECS on EC2** instead of Fargate, enabling high-density multiplayer servers with granular control over the compute resources used for each game, resulting in a **43% reduction in per-game hosting costs**.
- Stored game networking info into a **Redis** caching layer, enabling a custom reverse proxy to connect users in **<2 seconds**.
- Implemented container-level monitoring to automatically shut unhealthy instances and allow a usage-based pricing model.
- Engineered a **Step Functions**-orchestrated provisioning pipeline to reliably build, deploy, and terminate per-game server instances, achieving **~40 seconds** between request time to live server.

Fitted | Typescript, Next.js, Tailwind, PostgreSQL, AWS (S3, CloudFront), OpenAI Embedding

April 2025 – April 2025

Led a team of 4 to win AI track (1 of 50) at HackUSF; shipped full-stack web app in 24 hours.

- Built an **AI** outfit generation webapp with **TypeScript** and **React** with a type-safe tRPC API layer and **PostgreSQL** storage.
- Formulated a clothing embedding model to capture style, appearance, and functionality, utilizing a cosine-similarity search layer over embedded clothing vectors, enabling a **sub-500ms semantic outfit matching** system across wardrobe images.
- Deployed **AWS S3 and CloudFront** caching for scalable asset storage and delivery to eliminate clothing item load times.
- Established a **Git** branching strategy for a 4-person team and onboarded teammates to use a new **TypeScript**-based stack.

Imgscrpr | Typescript, MongoDB, Redis, REST API

October 2021 - February 2022

Created and monetized a Discord bot added by 100+ servers, converting several users to a paid tier.

- Used **Typescript** and **Redis** to implement a rate-limit aware caching strategy, remaining compliant with Reddit API policies while supporting **100+ concurrent servers**, along with providing a **<2 second response time** to Discord chats.
- Instituted a feature gating and Patreon billing pipeline, converting free servers using a **recurring-revenue** pricing model.
- Scaled to **maximum server-cap** for unverified bots in one week, shut down after hitting third-party data policy constraint.

SKILLS

Languages: TypeScript, JavaScript, Java, Python, SQL, Bash, C/C++, HTML/CSS

Application Development: React, Next.js, Tailwind, Node.js (Express), Flask, tRPC

Infrastructure & DevOps: AWS (EC2, ECS, Lambda, S3, Step Functions, VPC, CloudFront), CI/CD, Docker

Data Systems: PostgreSQL, Redis, MongoDB, Drizzle ORM, Prisma

WORK EXPERIENCE

Publix Supermarkets

Customer Service Staff

Largo, FL / Tampa, FL

March 2022 – Present

Directed checkout floor operations for teams of 10+ associates during periods of high traffic, optimizing for customer flow.

Entrusted with end-of-day reconciliation and high-value (\$3k+) transactions, maintaining financial accuracy.